BDA CAPSTONE PROJECT

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Basics of Data Analytics

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INTRODUCTION

Patron is a Fast-Growing specialty coffee and café brand with multiple outlets across India

Loyalty Rewards Campaign was introduced around Quarter 4 of 2023 to encourage repeat visits and increase customer spending

In this report, we will evaluate:

Overall Business Performance in Q2 of 2025

Loyalty Rewards Campaign's Impact



THE PROCESS

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DATA PROVIDED	Description
- Customer_Information.csv	Customer Profiles
- Menu_Information.csv	Menu of Café
- Loyalty_Campaign_Information.csv	Year End Campaign Records
- Order_Information.xslx	All orders placed in Q2 2025

DATA CLEANING

CHECKING FOR DUPLICATE DATASET ENTRIES

We verified that no duplicate records are present in the following tables:

1. Customer Information Table
2. Order Information Table

CHECKING FOR MISSING VALUES

We verified that no missing values were found in both tables:

1. Customer Information Table
2. Order Information Table

DATA CONSISTENCY

The datatypes used are real integers, characters and strings.
These datatypes are standard and can be dealt with without issues.

Hence the Data Cleaning Step Has Ended.

PROOF OF APPLICATION

```
print("Customer_Information.csv")
display(customers[customers.duplicated()])
time.sleep(2)
clear_output(wait=True)
print("Order_Information.xlsx")
display(orders[orders.duplicated()])
time.sleep(2)
clear_output(wait=True)
print("Hence, \nDuplicate Customer: ", customers.duplicated().sum(), "\nOrders Customer: ",
orders.duplicated().sum())
customers.drop_duplicates(inplace=True)
orders.drop_duplicates(inplace=True)
```

```
Hence,
Duplicate Customer:  0
Orders Customer:  0
```

DESCRIPTIVE STATISTICS

INFERENCE TABLE

Metric	Mean	Median (50th)	Std Deviation	Minimum	Maximum
Unit Price (INR)	290.92	240.00	208.89	80.00	1100.00
Quantity Ordered	2.05	2.00	1.60	1.00	32.00
Total Amount (INR)	571.60	420.00	619.75	68.00	9240.00
Customer Rating	3.05	3.00	1.42	1.00	5.00
Discount Applied (%)	3.56%	0.00%	8.16%	0.00%	50.00%

PROOF OF APPLICATION

```
#DESCRIPTIVE STATISTICS
dstats_orders=orders[['Unit_Price_INR', 'Quantity', 'Total_Amount_INR', 'Customer_Rating',
'Discount_%']].describe()
dstats_orders
```

	Unit_Price_INR	Quantity	Total_Amount_INR	Customer_Rating	Discount_%
count	1711.000000	1711.00000	1709.000000	1272.000000	1711.000000
mean	290.923437	2.05377	571.600410	3.047956	3.556108
std	208.892850	1.60428	619.746654	1.424213	8.156769
min	80.000000	1.00000	68.000000	1.000000	0.000000
25%	180.000000	1.00000	255.000000	2.000000	0.000000
50%	240.000000	2.00000	420.000000	3.000000	0.000000
75%	320.000000	3.00000	660.000000	4.000000	0.000000
max	1100.000000	32.00000	9240.000000	5.000000	50.000000

CUSTOMER ANALYSIS

PROBLEM 1: UNIQUE CUSTOMERS

The number of **unique customers is 200**

```
unique_customers = orders['Customer_ID'].nunique()
print(f"Unique customers ordered in Q2 2025: {unique_customers}")
```

PROBLEM 2: DISTRIBUTION

Distribution as per Region / Cities / Tier, we find the following inferences:

Total Revenue

```
Region Dist: {'West': 477, 'East': 438, 'South': 412, 'North': 384}
City Dist: {'Kolkata': 170, 'Bengaluru': 136, 'Jaipur': 118, 'Surat': 116, 'Nagpur': 112}
Tier Dist: {'Standard': 615, 'Silver': 588, 'Gold': 269, 'Platinum': 174}
Age Group Dist: {'55+': 441, '46-55': 343, '26-35': 336, '36-45': 269, '18-25': 247, '<18': 15}
```

Proof of findings are as follows:

Distribution as per District / City / Tier

```
print("Region Dist:", merge['Region'].value_counts().to_dict())
print("City Dist:", merge['City'].value_counts().head(5).to_dict())
print("Tier Dist:",
merge['Membership_Tier'].value_counts().to_dict())
```

Distribution as per Age

```
bins = [0, 18, 25, 35, 45, 55, 100]
labels = ['<18', '18-25', '26-35', '36-45', '46-55', '55+']
merge['Age_Group'] = pd.cut(merge['Age'], bins=bins, labels=labels)
print("Age Group Dist:",
merge['Age_Group'].value_counts().to_dict())
```

PROBLEM 3: AVERAGE SPEND

The average spending per customer for the quarter was INR 4,884.33p.

```
avg_spend_per_cust = merge.groupby('Customer_ID')['Total_Amount_INR'].sum().mean()
print(f"Average spend per customer in Q2: INR {avg_spend_per_cust:.2f}")
```

SALES & MENU ANALYSIS

PROBLEM 1: MENU PERFORMANCE

The **Top Items** and **Top Categories** are as follows:

Top Items & Categories

```
Top Items: {'Oat Bowl': 61, 'Espresso': 60, 'Avocado Toast': 60, 'Protein Smoothie': 59, 'Almond Danish': 58}
Top Categories: {'Beverages': 479, 'Meals': 421, 'Bakery': 396, 'Specialty': 267, 'Merchandise': 148}
```

Menu Performance

```
top_items =
orders['Product_Name'].value_counts().head(5).to_dict()
top_customers =
orders['Category'].value_counts().head(5).to_dict()
print("Top Items:", top_items)
print("Top Categories:", top_categories)
```

PROBLEM 2: TOTAL REVENUE

Total Revenue generated during Quarter 2 of 2025 is: **9,76,865.00 INR**

Total Revenue

```
total_revenue = orders['Total_Amount_INR'].sum()
print(f"Total Revenue Q2 2025: INR {total_revenue:.2f}")
```

PROBLEM 3: ORDER TYPE AND PAYMENT TRENDS

Maximum Orders are Dine-in and Take-away and Delivery being the least used comparatively

Order Type & Payment Trends

```
Sales by Order Type: {'Delivery': 231082.06, 'Dine-In': 433913.57, 'Takeaway': 311869.47}
Sales by Payment Method: {'Cash': 197676.18, 'Credit Card': 178434.87, 'Debit Card': 214294.47, 'UPI': 188184.47, 'Wallet': 198275.11}
Avg Spend per Tier: {'Gold': 494.7408550185874, 'Platinum': 609.8762643678161, 'Silver': 591.7027891156463, 'Standard': 590.0507340946166}
```

LOYALTY CAMPAIGN ANALYSIS

PROBLEM 1: LOYAL CUSTOMERS BUY-BACK

Customers who received the loyalty perk and actually ordered are 140.

OUTPUT

```
Customers who received a reward and ordered: 140
```

Loyal Buy-Back Customers

```
orders_loyalty = pd.merge(merge, loyalty, on='Customer_ID', how='left')

recipients = loyalty['Customer_ID'].unique()
recipients_ordered = orders[orders['Customer_ID'].isin(recipients)]
['Customer_ID'].nunique()
print(f"Customers who received a reward and ordered: {recipients_ordered}")
```

PROBLEM 2: SPEND COMPARISON

The Spending of Recipients increased when compared to those of non-recipients

OUTPUT

```
Avg Spend - Recipients: INR 5599.25, Non-Recipients: INR 3216.17
```

Spend Comparison

```
merge['Is_Recipient'] = merge['Customer_ID'].isin(recipients)
spend_comparison = merge.groupby(['Customer_ID', 'Is_Recipient'])
['Total_Amount_INR'].sum().reset_index()
avg_spend_recip = spend_comparison[spend_comparison['Is_Recipient'] ==
True]['Total_Amount_INR'].mean()
avg_spend_nonrecip = spend_comparison[spend_comparison['Is_Recipient'] ==
False]['Total_Amount_INR'].mean()
print(f"Avg Spend - Recipients: INR {avg_spend_recip:.2f}, Non-Recipients:
INR {avg_spend_nonrecip:.2f}")
```

PROBLEM 3: EFFECTIVE REWARD OFFER

Most Effective Driving Sales Volume & Frequency:
The most profitable for the company was **'10% off next visit'**
The most marginal for the company was **'Free Beverage Coupon'** due to its high average spending when compared to the 10% off next visit coupon.
Overall, the most profitable was "10% off next visit coupon"

LOYALTY CAMPAIGN ANALYSIS

PROBLEM 4: EFFECTIVE REWARD OFFER (THINKING)

Spend vs Sales

```
loyalty_spend =
pd.merge(spend_comparison[spend_comparison['Is_Recipient']==True], loyalty,
on='Customer_ID')
print("Avg Spend by Reward Type:", loyalty_spend.groupby('Reward_Offer')
['Total_Amount_INR'].mean().sort_values(ascending=False).to_dict())
print("Total Sales by Reward Type:", loyalty_spend.groupby('Reward_Offer')
['Total_Amount_INR'].sum().sort_values(ascending=False).to_dict())
```

PROBLEM 5: OVERALL REDEMPTION PATTERN

Most Redeemed coupon was **“10% Off Next Visit”** and the least drawn coupon was **“Free Patry Coupon”.**
This was predictable by human extinct because 10% off next visit had better interpretations in average spending and total spending.

OUTPUT

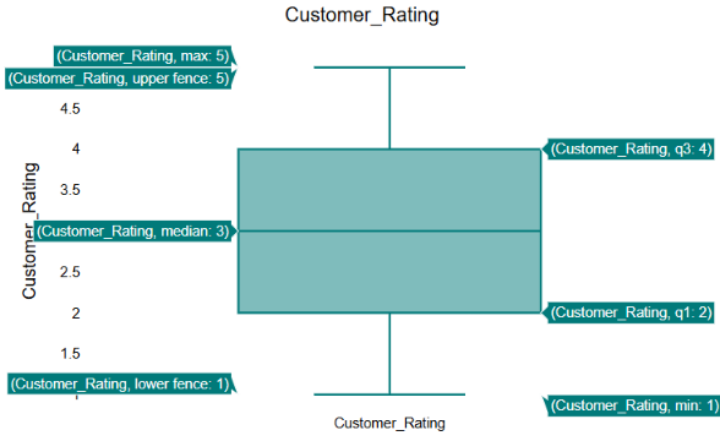
```
Redemption Pattern (Reward_Applied counts in Orders): {'10% Off Next
Visit': 245, 'Double Loyalty Points': 130, '15% Off Total Bill': 119, 'Free
Beverage Coupon': 118, 'Buy 1 Get 1 Beverage': 92, 'Free Pastry Coupon':
63}
```

Reward Applied

```
print("Redemption Pattern (Reward_Applied counts in Orders):",
orders['Reward_Applied'].value_counts().to_dict())
```

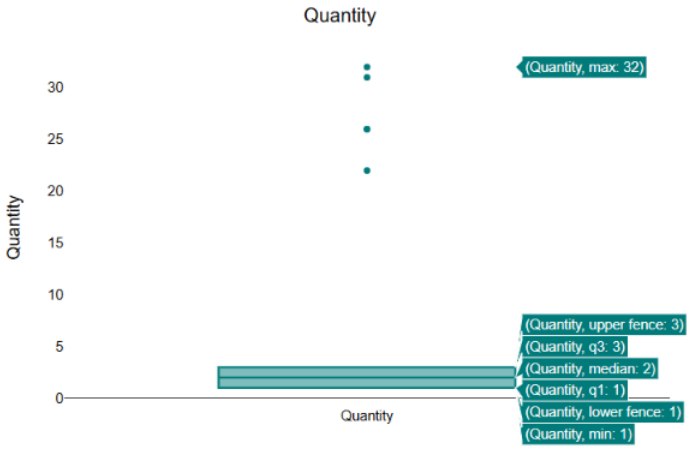
ADVANCED ANALYSIS

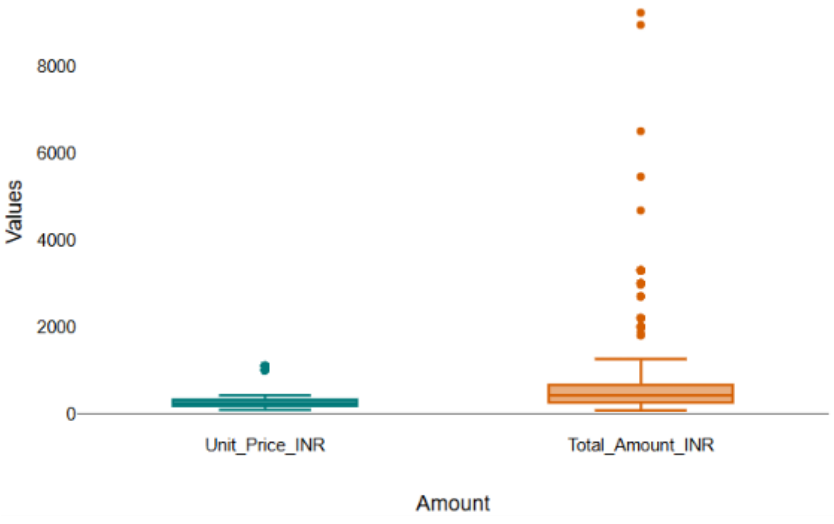
PROBLEM 3: BOX PLOTS FOR KEY METRICS



We find that the median rating is three, which means the opinion of population upon the café is moderate, neither good nor bad.

From this plot,
We can see that there are many outliners present, this is since the quantity ordered with a family might be overwhelmingly high. This might cause the creation of these Outliners.



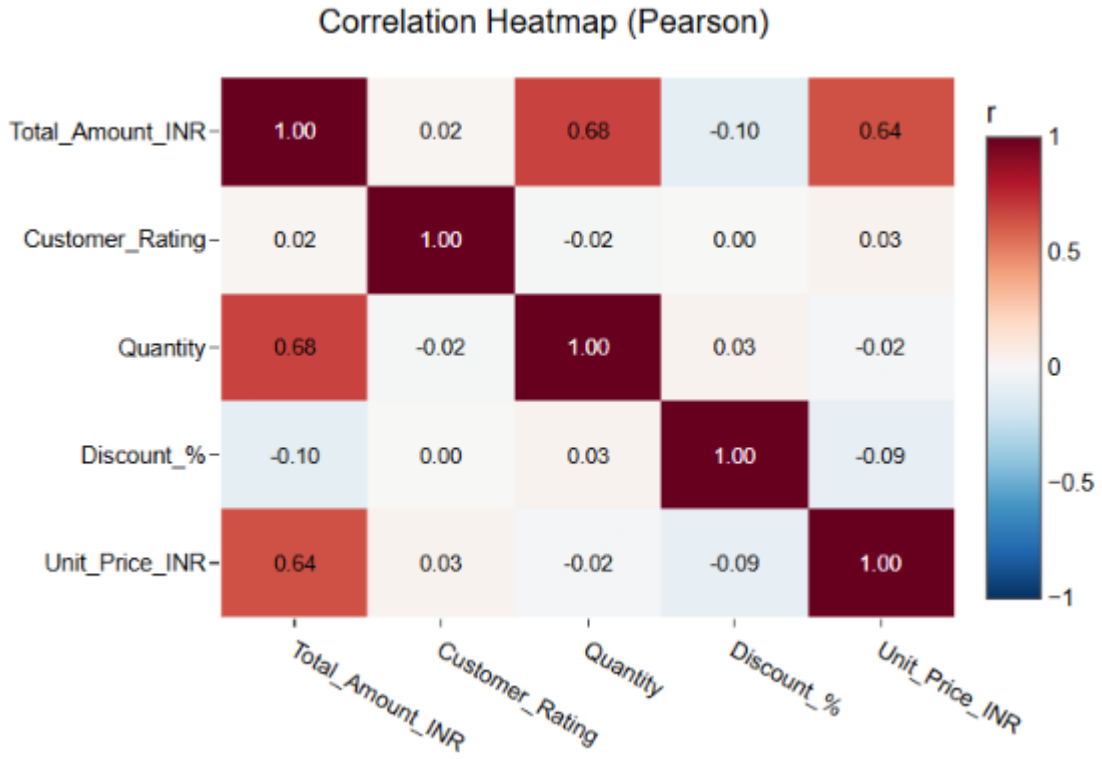


Through this, we can be confident that many people come to do meetups,etc

ADVANCED ANALYSIS

PROBLEM 3: HEAT MAP FOR CORRELATION

We plotted a heat-map to find the correlation between certain variables.



Here,

- Total amount is positively correlated with the quantity and unit price, which is indeed **true,**
- Total amount also has a negative correlation with discount percentage,
- Customer Rating has no correlation with almost every other variable until now.

INTERPRETATION

- The end of the year campaign has a **overall positive impact on the sales**
- The campaign was **highly successful in driving customers addictiveness**
- **We see that most people that visit this café above above 40+** (Senior Citizens)
- We see that the café is good at capturing large orders
- Most of it's revenue comes from **beverages**

ADVICE

- By the **10% off next visit** coupon, the café successfully manages to bring back customers and convert them into regular customers to earn profits
- The café shall remove the pastry coupon and scale the top 2 best performing coupons (10% off next visit / Free Pastry)

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